

CMI Entrance Examination

Mock Test - Part A (Solutions)

Solution to Question 1

- 1.1** $\frac{3\pi}{16}$. The integral $I_n = \int_0^{\pi/2} \sin^n(x) dx$ is evaluated using Wallis' formulas. For $n = 4$, $I_4 = \frac{4-1}{4} \cdot \frac{4-3}{4-2} \cdot \frac{\pi}{2} = \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{16}$.
- 1.2** **False.** The sequence is given by $A_n = nI_nI_{n-1}$. Using the properties of Wallis' integrals, $nI_nI_{n-1} = \frac{\pi}{2}$ for all integers $n \geq 1$. Since it evaluates to a constant value, the sequence is constant, not strictly decreasing.
- 1.3** $\sqrt{\frac{\pi}{2}}$. Using Wallis' product and Stirling's approximation, it is a known asymptotic limit that $I_n \sim \sqrt{\frac{\pi}{2n}}$ as $n \rightarrow \infty$. Thus, $\lim_{n \rightarrow \infty} \sqrt{n}I_n = \sqrt{\frac{\pi}{2}}$.
- 1.4** $1 - \frac{\pi}{2}$. The sum $S = \sum_{n=1}^{\infty} (-1)^n I_n$ can be rewritten by moving the sum inside the integral:

$$S = \int_0^{\pi/2} \sum_{n=1}^{\infty} (-\sin(x))^n dx = \int_0^{\pi/2} \frac{-\sin(x)}{1 + \sin(x)} dx$$

This simplifies to $\int_0^{\pi/2} \left(\frac{1}{1+\sin(x)} - 1 \right) dx$. Evaluating $\int \frac{1}{1+\sin(x)} dx$ yields $\tan(x) - \sec(x)$, which evaluates to 1 over the bounds $[0, \pi/2]$. Subtracting $\int_0^{\pi/2} 1 dx = \frac{\pi}{2}$ gives the final answer.

Solution to Question 2

- 2.1** **False.** The sequence is $a_n = 2^{2n+1} - 2^{n+1} + 1$. For $n = 2$, $a_2 = 2^5 - 2^3 + 1 = 32 - 8 + 1 = 25$, which is not a prime number.
- 2.2** **1.** By rewriting the sequence, we find that $a_n \equiv 0 \pmod{5}$ when $n \equiv 1$ or $2 \pmod{4}$. This means a_{10} is divisible by 5, but a_{11} is not. Factoring the recursion reveals no shared divisors for consecutive terms other than potential factors of 5, which doesn't divide a_{11} . Thus, they are coprime.
- 2.3** **50.** As established above, $a_n \equiv 0 \pmod{5}$ precisely when $n \equiv 1 \pmod{4}$ or $n \equiv 2 \pmod{4}$. Within the first 100 positive integers, exactly half meet this condition.
- 2.4** **5.** We evaluate the terms modulo 7. Since $2^3 \equiv 1 \pmod{7}$, we look at the exponent modulo 3. $2026 \equiv 1 \pmod{3}$, so $2026 \times 2 + 1 = 4053 \equiv 0 \pmod{3}$, giving $2^{4053} \equiv 1 \pmod{7}$. For the middle term, $2026 + 1 = 2027 \equiv 2 \pmod{3}$, giving $2^{2027} \equiv 4 \pmod{7}$. Thus, $a_{2026} \equiv 1 - 4 + 1 \equiv -2 \equiv 5 \pmod{7}$.

Solution to Question 3

- 3.1 5.** The partitions of 8 into exactly 3 strictly positive parts are: $(6, 1, 1)$, $(5, 2, 1)$, $(4, 3, 1)$, $(4, 2, 2)$, and $(3, 3, 2)$.
- 3.2 True.** This is mathematically true because these two sets of partitions are conjugates of each other, which can be visualized by transposing their Ferrers diagrams.
- 3.3 7.** The formal power series given is the generating function for the number of integer partitions into parts of sizes 1, 2, and 3. The coefficient of x^6 is the number of ways to partition 6 using only 1s, 2s, and 3s. There are 7 such ways: $(3, 3)$, $(3, 2, 1)$, $(3, 1, 1, 1)$, $(2, 2, 2)$, $(2, 2, 1, 1)$, $(2, 1, 1, 1, 1)$, and $(1, 1, 1, 1, 1, 1)$.
- 3.4 4.** The distinct odd parts summing to 15 are: (15) , $(11, 3, 1)$, $(9, 5, 1)$, and $(7, 5, 3)$.

Solution to Question 4

- 4.1 True.** In an acute-angled triangle, simple trigonometry on the circumcircle proves that the perpendicular distance from the circumcenter to the side BC is exactly $R \cos A$.
- 4.2 True.** This is a standard trigonometric distance identity for the Euler line relating the orthocenter H and circumcenter O .
- 4.3 $\frac{\sqrt{310}}{8}$.** Using the area $K = 6\sqrt{6}$ and circumradius $R = \frac{abc}{4K} = \frac{35}{4\sqrt{6}}$, we apply the formula $OH^2 = 9R^2 - (a^2 + b^2 + c^2)$.
- $$OH^2 = 9 \left(\frac{1225}{96} \right) - (25 + 36 + 49) = \frac{3675}{32} - 110 = \frac{155}{32}$$
- Taking the square root simplifies to $\frac{\sqrt{310}}{8}$.
- 4.4 $\frac{1}{2}$.** By Euler's inequality, $R \geq 2r$ for any valid triangle, which rearranges to $\frac{r}{R} \leq \frac{1}{2}$. The strict maximum value occurs when the triangle is equilateral.

Solution to Question 5

- 5.1 3.** Analyzing the local extrema of $P(x) = x^5 - 5x^3 + 4x - 2$, the derivative $P'(x) = 5x^4 - 15x^2 + 4$ yields roots that represent two local maxima and two local minima. Evaluating $P(x)$ at these critical points demonstrates exactly 3 real crossings over the x -axis.
- 5.2 34.** Using Newton's sums on the polynomial where elementary symmetric polynomials are $e_1 = 0, e_2 = -5, e_3 = 0, e_4 = 4$, the sum of the fourth powers is calculated recursively as $S_4 = e_1 S_3 - e_2 S_2 + e_3 S_1 - 4e_4 = 0 - (-5)(10) + 0 - 4(4) = 50 - 16 = 34$.
- 5.3 True.** By the rational root theorem, the only possible rational roots are $\pm 1, \pm 2$, none of which evaluate to 0. Verifying factorizations of degree 2 and 3 using undetermined coefficients yields no rational solutions, confirming it is irreducible over $\mathbb{Q}$.

- 5.4 4.** The product $\prod_{i=1}^5 (r_i^2 - 1)$ can be factored as $\prod (r_i - 1) \prod (r_i + 1)$. Based on the polynomial evaluation, this maps to $P(1) \times P(-1)$. Since $P(1) = -2$ and $P(-1) = -2$, the final product is exactly 4.

Solution to Question 6

- 6.1 0.** The sum of the m -th powers of the n -th roots of unity is 0 whenever m is not a multiple of n . Since 7 is not a multiple of 11, the sum strictly evaluates to 0.
- 6.2 11.** The roots satisfy $\prod_{k=2}^{11} (z - z_k) = z^{10} + z^9 + \cdots + 1$. Evaluating this geometric product at $z_1 = 1$ gives $1 + 1 + \cdots + 1 = 11$.
- 6.3 False.** For three roots on the unit circle to form an equilateral triangle, the angular distance between them must be exactly $\frac{2\pi}{3}$. Because 3 does not divide 11, this configuration is impossible.
- 6.4** $\sin\left(\frac{8\pi}{11}\right) + \frac{1}{2} \sin\left(\frac{6\pi}{11}\right)$. To maximize the area, the three chosen vertices must be as evenly spaced as possible around the 11-gon. This corresponds to angular steps of 4, 4, and 3 times $\frac{2\pi}{11}$. Substituting these angles into the area formula for an inscribed polygon $\frac{1}{2} R^2 \sum \sin(\theta_i)$ gives this maximum expression.

Solution to Question 7

- 7.1 10.** This is a standard sum of sines in arithmetic progression. The series $f(x) = \sin(x) + \sin(3x) + \cdots + \sin(19x)$ condenses to $\frac{\sin^2(10x)}{\sin(x)}$, making $A = 10$.
- 7.2** $\csc\left(\frac{\pi}{20}\right)$. Substituting $x = \frac{\pi}{20}$ into the closed-form expression gives $\frac{\sin^2(\pi/2)}{\sin(\pi/20)} = \frac{1}{\sin(\pi/20)}$, which is exactly $\csc\left(\frac{\pi}{20}\right)$.
- 7.3 True.** The roots in the interval $(0, \pi)$ occur when $\sin(10x) = 0$ but $\sin(x) \neq 0$. This yields $10x = k\pi \implies x = \frac{k\pi}{10}$ for $k \in \{1, 2, \dots, 9\}$, which gives exactly 9 distinct solutions.
- 7.4** $\frac{\pi}{2}$. The integral is $\int_0^\pi \sin^2(10x) dx$. Using the half-angle substitution $\frac{1 - \cos(20x)}{2}$, the cosine term integrates to 0 over a full period, leaving exactly $\frac{\pi}{2}$.

Solution to Question 8

- 8.1 5.** Let $D_n = \det(M_n)$. Expanding the determinant by minors gives the recurrence relation $D_n = 2D_{n-1} - D_{n-2}$ with $D_1 = 2$ and $D_2 = 3$. This generates the sequence $D_n = n + 1$. For $n = 4$, the determinant is 5.
- 8.2 True.** As proven by the induction in 8.1, the determinant of this specific tridiagonal matrix is always exactly $n + 1$.
- 8.3 6.** Multiplying M_5 by the column vector $(1, 2, 3, 4, 5)^T$ performs the operation $-x_{i-1} + 2x_i - x_{i+1}$. This evaluates to 0 for the first four rows and strictly 6 for the last row.

- 8.4 True.** By Sylvester's criterion, a matrix is strictly positive definite if all its leading principal minors are positive. Since the minor $D_k = k + 1 > 0$ for all $k \geq 1$, the matrix is strictly positive definite.

Solution to Question 9

- 9.1 True.** The recursion $x_{n+1} = \frac{1}{2} \left(x_n + \frac{7}{x_n} \right)$ is Newton's method for finding $\sqrt{7}$. Since the initial value $x_1 = 3 > \sqrt{7}$, every subsequent term over-approximates $\sqrt{7}$ and the sequence strictly decreases toward the root.
- 9.2 $\sqrt{7}$.** Setting the limit $L = x_{n+1} = x_n$ yields $L = \frac{1}{2} \left(L + \frac{7}{L} \right)$, which resolves exactly to $L = \sqrt{7}$.
- 9.3 2.** Substituting the recursive definition of x_{n+1} into the fraction y_{n+1} mathematically simplifies to $\left(\frac{x_n - \sqrt{7}}{x_n + \sqrt{7}} \right)^2$, thereby proving $y_{n+1} = y_n^2$, making $k = 2$.
- 9.4 True.** Newton's method exhibits quadratic convergence. The terms $x_n - \sqrt{7}$ decrease doubly-exponentially (proportional to $y_1^{2^{n-1}}$), meaning the infinite series of these terms converges incredibly fast.

Solution to Question 10

- 10.1 8.** On an 8×8 grid, a rook commands its entire row and column. Placing 8 rooks along the main diagonal (or any configuration with one in each row and column) reaches the maximum non-attacking limit of 8.
- 10.2 False.** Using a 3-color scheme where each square (i, j) is assigned a color based on $(i + j) \pmod{3}$, a straight 1×3 tromino will always cover exactly one square of each color. A standard 8×8 grid lacks color parity for this covering unless the removed tile is geometrically restricted to specific coordinates (row 3 or 6, and column 3 or 6). The center squares do not satisfy this.
- 10.3 1296.** An 8×8 grid consists of 9 vertical lines and 9 horizontal lines. A rectangle is formed by choosing any 2 distinct horizontal lines ($\binom{9}{2} = 36$ ways) and any 2 distinct vertical lines ($\binom{9}{2} = 36$ ways). The total is exactly $36 \times 36 = 1296$ rectangles.
- 10.4 False.** This is the mutilated chessboard problem. A standard domino covers exactly one black and one white square. Diagonally opposite corners share the same color. Removing them leaves an unequal number of colored squares (32 of one, 30 of the other), making a perfect 1×2 covering mathematically impossible.